A Planar Air Levitated Electrostatic Actuator System

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Abstract

A micromotor capable of moving multiple objects in three dimensions has been fabricated. The fixed surface of the motor, fabricated on a silicon wafer, contains air nozzles which levitate rigid platforms, as well as conductive plates which generate an electric field to apply forces to the platforms. Initial testing has shown the air nozzles form a stable very low friction bearing, and the motor is capable of several hundred microns of motion in two dimensions with fields generated by a 2 volt potential. Applications of this motor include a microrobotic work cell.

Background

Microfabricated sensors and actuators are abundant in laboratories and in industry. These devices present a unique opportunity for the development of robotic work cells composed of a combination of sensors and actuators on one or more platforms moving in a chip sized work space. The motion of these sensor/actuator platforms should ideally be generated by devices on the same scale as the platforms themselves. This motivates the study of micromotors capable of generating motion in two or more dimensions.

Work in the field of micro-motors has shown that frictional forces play a dominant role in the dynamics of micro-machines [1]. Recently much effort as been directed at removing friction entirely [2], or using it as an integral part of the functionality of the motor [3][4]. Unfortunately most of these motors generate either oscillatory or rotary motion which can not be used directly for translation of sensor/actuator platforms.

A relatively unexplored area involves minimizing friction by ensuring that moving parts do not come into contact. This has been done using superconducting electromagnetic levitation [5] and a combination of free fall and magnetic clamping [6]. These systems have an advantage in that they are capable of linear motion in one or more directions. Our approach is similar to these. In order to minimize friction, the moving elements of our design ride on a cushion of air supplied through nozzles in the fixed surface of the motor. These moving parts, or platforms, are then actuated by electrostatic fields generated by conductors in the surface of the motor (Figure 1).

Design

Our goal was to levitate and actuate platforms roughly 1 mm square. To generate a more uniform air cushion we chose the nozzle spacing (100μm) to be small compared to the dimension of the platform. The nozzle dimension used was 10μm . Lacking the technology to fabricate 10μm nozzles through a full wafer thickness, we decided to create nozzles through membranes and supply pressure behind the membrane. Given expected supply pressures of 100-1000 pascals and the need to have a very flat surface (for the bearing to function) it became obvious that the membranes would have to either be very thick (greater than 10μm) or very narrow (less than 500 μm).

Due to processing constraints the narrow membrane approach was chosen. Each bearing surface is made up of 20 parallel low stress silicon nitride membranes, each membrane is 9 mm long by 130 μm wide and 1 μm thick . The membranes cover air supply trenches beneath them, and air is introduced to the underside of the membrane via front surface supply holes on one side of the membranes (Figure 4).

The fabrication of the air bearings is straightforward. An oxide layer is patterned to define the future membrane locations. Silicon nitride is grown on top of the oxide and the air nozzles are patterned through the nitride, contacting the oxide layer. The oxide is etched in HF, freeing the nitride membranes. The resulting air supply channels are not sufficiently large, so the structure is then immersed in KOH to bulk etch the silicon under the membranes [7][8].

In one version of the process, after the membranes are created, aluminum is sputtered and patterned to form electrodes. Exposed electrodes can lead to problems such as shorting, and the lack of an insulating cover for the electrodes allows vertical attractive forces on the platforms to become extremely high, thereby making unwanted clamping more common. To avoid this problem, polysilicon electrodes are grown and patterned after the first nitride deposition. A second layer of nitride is then deposited to insulate the electrodes before the air nozzles are patterned and the membranes freed (Figure 2).

Air Bearing

Figure 7 shows a cross section of a 1 mm platform floating on the bearing. P_i represents the pressure above the ith nozzle, Q_i the flow between nozzles, Q_{Si} the flow through the ith nozzle. P_a is atmospheric pressure (gauge pressure 0), and P_S is the pressure supplied to the underside of the membranes (assumed constant). Assuming fully developed laminar incompressible flow (i.e. Reynolds number << 1400 and Mach number << 1), the volumetric flow rate between supply holes, Q_i, will be a linear function of the pressure drop between the holes, similarly the flow through the nozzles will be proportional to the pressure drop across the nozzle. These approximations ignore entrance and exit effects, which are typically non-linear. These relations are expressed as

$$P_i - P_{i+1} = R_i Q_i \quad (1)$$

$$P_S - P_i = R_S Q_{Si} \quad (2)$$

The flow resistances R_i will be functions of the altitude, h_i, of the platform between the ith and the $(i+1)^{th}$ nozzles. R_S is constant for a given nozzle size. For incompressible viscous flow in a finite slot the resistance is

$$R_i = \frac{12\mu L}{w h_i^3} \quad (3)$$

where L and w are the length and width of the channel, respectively, and μ is the viscosity of air ($1.78 \times 10^{-5} Kg/m \cdot s$) [9]. For a platform floating at 20μm R_I is $2.7 \times 10^{10} Pa \cdot s \cdot m^{-3}$.

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The nozzle resistance can be approximated by the resistance of an inscribed circular nozzle,

$$R_S = \frac{8\mu L}{r^4} \tag{4}$$

where L is the thickness of the membrane. For a 10μm nozzle this gives a theoretical nozzle resistance of $7.3 \times 10^{10} Pa \cdot s \cdot m^{-3}$. The measured nozzle resistance at 500 pascals applied pressure was roughly $1.9 \times 10^{11} Pa \cdot s/m^3$, in reasonable agreement with the theoretical value given that entrance effects on the flow were ignored in the theoretical calculation.

Given these resistance values the pressure distribution under the platform can be found and from it the load carrying capacity of the bearing. The pressure above the ith air nozzle is

$$P_i = \alpha_i P_{i-1} + \beta_i P_{i+1} + \gamma_i P_S \tag{5}$$

where P_S is the constant supply pressure (all pressures are gauge pressures), and

$$\alpha_i = R_s R_{i+1}/d_i \tag{6}$$
$$\beta_i = R_i R_s/d_i \tag{7}$$
$$\gamma_i = R_i R_{i+1}/d_i \tag{8}$$
$$d_i = R_i R_s + R_s R_{i+1} + R_i R_{i+1} \tag{9}$$

This set of linear equations can be solved for the P_i's. The average of these pressures is the theoretical load carrying capability of the bearing. The numbers calculated from this model are consistently three or four times the experimentally determined capability. This is at least partly due to the modelling assumption that the flow is only one dimensional. The model predicts flow rates of up to $3 \times 10^{-3} cm^3 s^{-1}$ under the edge of the platform (Q_0), with peak velocities of $30cm \cdot s^{-1}$. Flow through nozzles is predicted to be $10^{-4} cm^3 s^{-1}$ with average velocities less than $10m \cdot s^{-1}$.

Experimental values of air flow to the entire test chip are 5 to 10 $cm^3 s^{-1}$ at 500 pascals. With roughly 7000 nozzles per chip this means that average unobstructed flow through a single nozzle at a differential pressure of 500 pascals is $10^{-3} cm^3 s^{-1}$ with a flow rate of 10 $m \cdot s^{-1}$ at the nozzle, in rough agreement with the model. This velocity corresponds to a Mach number of .03, indicating that the flow is accurately modeled as incompressible. The Reynolds number for this flow is roughly 1, indicating that the flow is laminar and viscous. Another number of interest when working at this scale is the Knudsen number, the ratio of the mean free path of air (.064 μm at STP) to a characteristic dimension of the problem. With our current design, typical numbers for bearing thicknesses and nozzle dimensions are greater than 10μm , giving a Knudsen number less than .01, which indicates that the air can still be treated as a continuous fluid. Future bearings will likely have substantially smaller characteristic dimensions and will require somewhat more advanced analysis.

Electrostatics

Figure 8 shows a model of the capacitance between two electrodes with the platform above them. The assumptions of this model are that a conductive platform covers all of the left electrode, forming capacitor C_1, and only part of the right electrode, forming capacitor C_2. The total capacitance, C, between the two electrodes is C_0, the capacitance between the two electrodes with no platform present, in parallel with C_1 and C_2 in series. The total capacitance and the energy stored in the capacitor (as functions of the platform position) are

$$C(x,y) = C_0 + \frac{C_1(y)C_2(x,y)}{C_1(y) + C_2(x,y)} \tag{10}$$
$$U(x,y) = \frac{1}{2} C(x,y) V^2 \tag{11}$$

The force on the platform due to an applied potential is then

$$F_x = \frac{1}{2} V^2 \frac{\partial C(x,y)}{\partial x} \tag{12}$$
$$F_y = \frac{1}{2} V^2 \frac{\partial C(x,y)}{\partial y} \tag{13}$$

Using this model and making the approximation that C_1 and C_2 are ideal parallel plate capacitors, we can estimate the forces on the platform as a function of position. Assuming the platform is a 1 mm square conductor and the electrodes are 80μm wide and extend the length of the platform, and the applied potential is 10 volts,

$$F_x = \frac{3.2 \times 10^{-10} \epsilon_0}{y(x + 8 \times 10^{-5})^2} \tag{14}$$
$$F_y = 4 \times 10^{-6} \epsilon_0 \frac{x}{y^2(x + 8 \times 10^{-5})} \tag{15}$$

This system was simulated with the platform centered on the right electrode and a potential of 10 volts [10]. Cushion thickness was varied between 1 and 20μm . The simulation results for forces on the platform are shown as function of float height in Figure 10. Note that the data has the expected shape: the vertical force appears to be roughly proportional to y^{-2}, and the lateral force roughly y^{-1}. The simulated vertical forces match those predicted in equation 15 to within 20%, but the predicted lateral forces are 2 to 5 times lower than their simulated counterparts. Clearly this is at least partly due to the nature of the approximations made in developing the model, but the two dimensional simulation results may not be substantially more accurate, given the three dimensional nature of the problem [11].

Simulations were also done of a dielectric platform under the same conditions. Figure 9 shows an example of the simulated equipotential lines around the platform. Simulated forces on a dielectric platform were roughly an order of magnitude lower than forces on the conductive platform.

Results

Air bearing

Our experimental setup is unfortunately rather imprecise. Currently a constriction valve and a water column barometer are used to regulate and measure pressure respectively. Despite this, we have successfully levitated a large variety of materials including 2μm thick nitride membrane fragments, a 1mm cube of silicon, aluminum foil (15 μm thick), and a scrap of binder paper.

The load carrying capability of the bearing was tested with several thicknesses of platforms from .25mm to 1mm. The platforms were stacks of 1 or more 1mm square pieces of silicon wafers, each 250μm thick. Air cushion thickness as a function of bearing pressure was determined by focusing on the top surface of the platform and recording change in focus position as a function of change in pressure. Results are shown in Figure 6. Pressures shown are recorded pressures at the gauge, not the actual pressure beneath the membranes. Membrane pressures may actually be less than one half the indicated pressure, due to flow resistance in the tube connecting the gauge with the test device.

Clearly bearing thicknesses of one or two mils are not optimal for this device, given that macro scale bearings typically have clearances in this range. Decreasing the nozzle size, or increasing the membrane thickness should improve bearing performance. Both of these would increase nozzle flow resistance, thereby decreasing the cushion thickness and increasing stability. The current bearings may yield better results if a more sophisticated pressure regulator is used.

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Motor

Three types of motion have been seen as a result of applied electric fields. Surface clamping and pinned rotation have been observed in addition to the expected horizontal translation. Surface clamping occurs when the attractive force of the electric field overcomes the repulsive force of the air bearing. Pinned rotation results from dirt between the bearing surface and the platform, causing the platform to in effect become a rotor pinned at the contaminant. Clamping and rotation have been observed with both dielectric and conductive platforms. Translation has been seen only with conductive platforms.

Currently only 1 pole translational motion has been seen; i.e., we have not used the motor as a multi-pole stepper motor, because of a lack of probing equipment. Translational motion has been observed at bearing pressures between 500 and 2000 pascals. Voltages between 2 and 15 volts typically cause platform motion of between 100 and 500 microns. This motion has been observed at frequencies between .5 and 10 hz, with the largest amplitude motion occurring around 1 or 2 hz. The amplitude decrease at lower frequencies may indicate that some charging of the platform is taking place. The actual electrostatic forces on the platform have not been determined, but a rough lower bound on the lateral component is 3×10^{-9} Newtons. Based on the simulations of electrostatic force this indicates a bearing cushion thickness of approximately 10μm , which is roughly what we expect the thickness to be.

Voltages above roughly 20 volts will overcome the air bearing levitation force and clamp the platform to the surface of the bearing. Once a platform has been clamped, only a small voltage is necessary to maintain surface contact, even if bearing pressure is increased to 10 kPa. This effect is due to the decrease in surface area of the platform exposed to pressurized air (just the air nozzle area, roughly 1% of the total area of the bearing surface).

Before clamping takes place, the vertical electrostatic force on the platform attracts it to the surface. Initial experimentation has indicated that the thickness of the air cushion decreases in a roughly linear manner as the vertical force is increased. Although we have not measured the effect of the electric field on the cushion thickness, we believe that the the vertical position of the platforms will be smoothly controllable up to the point of clamping.

Long Term Goals

Given the ability to move multiple platforms in the same work space, we envision a $1cm^2$ work surface with many different platforms on its perimeter, each platform with its own function. Perhaps the system could be used as a microassembly cell, with individual parts floated into the workspace and manipulated by two or more platforms acting together to form a gripper. Parts and platforms could be individually clamped to the surface while assembly or machining takes place. One advantage of this system is that the moving parts are easily replaced. If a platform outlives its usefulness it would simply be shuttled off the work surface and replaced by another. The idea of multiple robotic platforms cooperating in the same workspace has been succesfully engineered on the macro scale by Automatix, Inc., makers of Robotworld [12]. Robotworld has several platforms suspended from a one meter square steel sheet. The platforms ride on an air bearing, as in our system, but are actuated by linear electromagnetic stepper motors. Robotworld is typically used for pick and place operations such as IC board stuffing, but is capable of more advanced manipulation such as small parts assembly [13].

If lateral drive forces can be increased sufficiently it may be possible to have small power lines running to a platform, in which case essentially any microfabricated sensor or actuator could be placed on a mobile platform. Alternatively, it may be possible to have power available on the surface of the bearing, where platforms could be moved into position and then clamped to the surface to contact supply pads. A potential application of this idea would be a reconfigurable probe station for integrated circuit (IC) failure analysis. Platforms with microprobes or capacitive sensors could be maneuvered to their desired locations and clamped to the surface, then the system would be placed in contact with the IC to be tested.

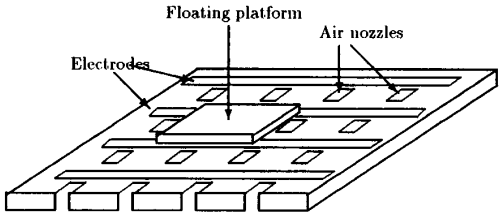


Figure 1: Illustration of a platform floating above the air nozzles and electrodes on the surface of the motor.

Conclusion

We have demonstrated a stable microfabricated air bearing capable of levitating objects between 2μm and 1mm thick with extremely low kinetic friction, and no static friction. In addition we have shown the compatibility of this bearing with an integrated electrostatic actuation system.

Testing has been limited to verification of the fundamental principles of operation, but experimental results indicate that a two dimensional linear electrostatic stepper motor should be realizable with this process. In addition, it should be possible to control the vertical motion of the platforms by varying either the supply voltage or bearing pressure.

Electrostatic forces on the platforms are currently on the order of 1 to 10 nN with an applied potential of 2 volts. If cushion thickness can be decreased to a few microns, it should be possible to attain forces several orders of magnitude higher than this, resulting in platform accelerations on the order of 1g. Research will be geared toward understanding and improving the air bearing in an effort to realize this goal. Likely directions of investigation are increasing nozzle flow resistance and using a backside air supply to improve pressure uniformity.

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Partial support of K. Pister provided by the Berkeley Sensor & Actuator Center, an NSF/Industry/University Cooperative Research Center. Many thanks to Tom Booth, Ben Costello, Leslie Field, Martin Lim, Carlos Mastrangelo, and Stuart Wenzel for their invaluable processing advice, assistance, and encouragement.

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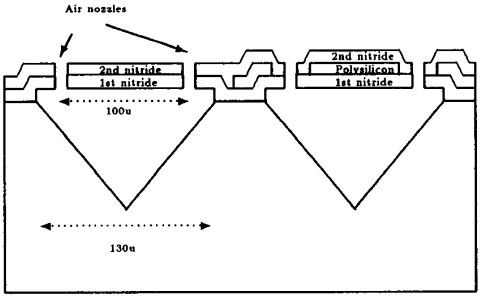


Figure 2: Cross section of the membranes and air channels showing air nozzles through the membranes and polysilicon electrodes embedded between the two layers of silicon nitride.

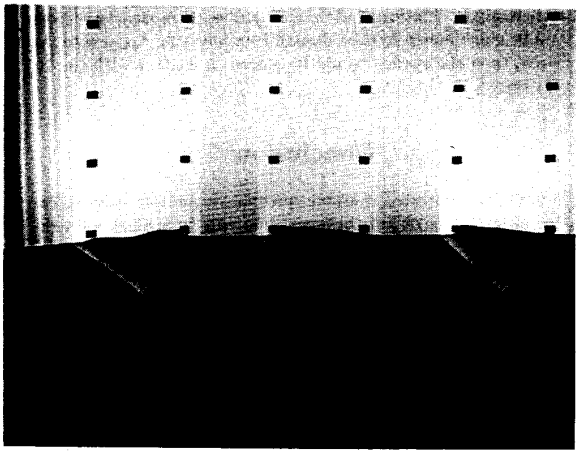


Figure 3: Scanning electron micrograph of membranes, air supply channels, and air nozzles. Faintly visible are buried polysilicon electrodes, 500μm square.

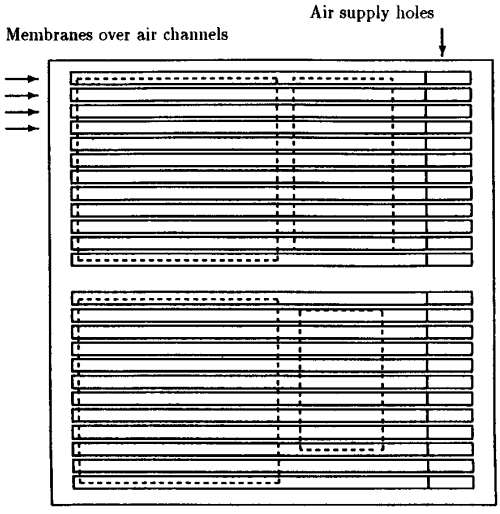


Figure 4: Basic outline of the test chip. Dashed boxes indicate areas with electrode test patterns.

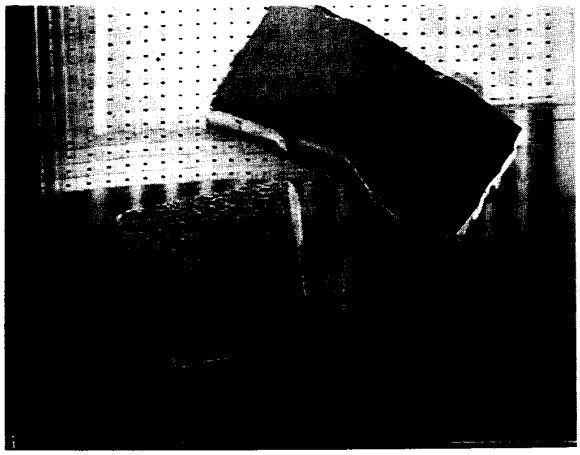


Figure 5: Detail micrograph showing two platforms at rest on the motor surface. Air nozzles are 100μm center to center, platforms are 120μm and 250μm thick.

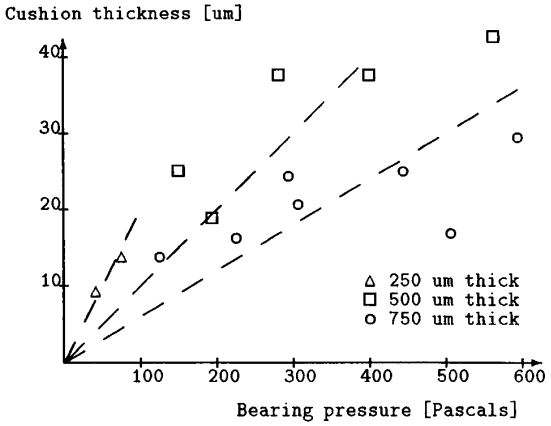


Figure 6: Bearing cushion height as a function of supply pressure. Data was taken with three platform thicknesses. All platforms were roughly 1 mm square.

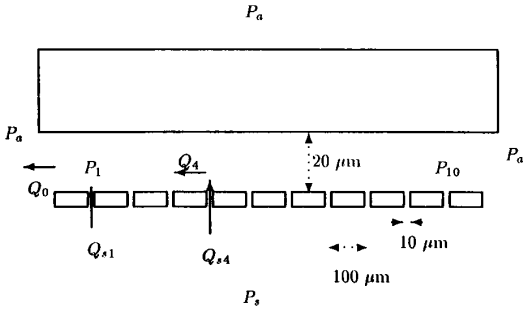


Figure 7: Cross section of platform above air nozzles showing pressures above (P_i) and flow through (Q_{si}) each nozzle, and flow under the platform (Q_i).

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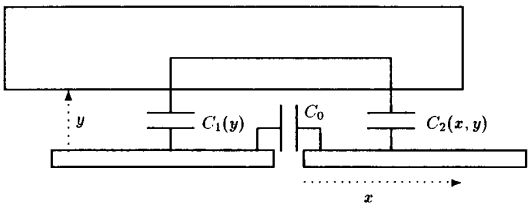


Figure 8: A model of the variable capacitance between electrodes

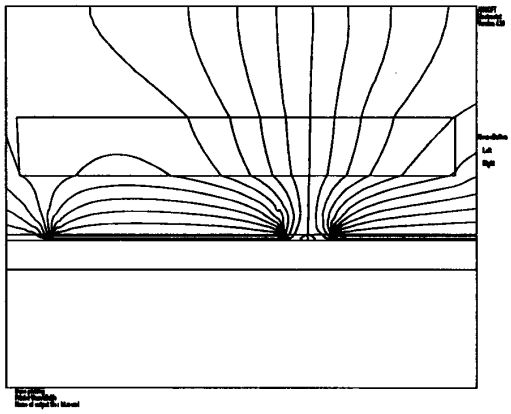


Figure 9: Simulation of device cross section showing equipotential lines. Platform is an uncharged dielectric, electrodes are at -5V and +5V above a grounded substrate.

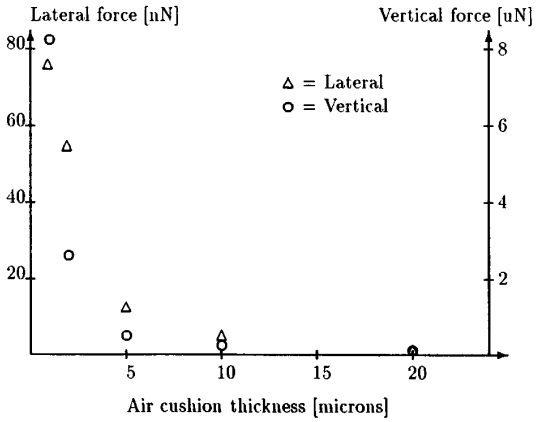


Figure 10: Simulated electrostatic forces on the platform as a function of cushion thickness.

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